

Yichao Jin, Ph.D.

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Specialization: Behavioral AI / Health Economics / Behavioral Economics / Discrete Choice Experiments

Education

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| University of Texas at Dallas | May 2026 |
| Ph.D. in Public Policy and Political Economy | |
| <ul style="list-style-type: none">- Dissertation: “Intertemporal Choice in Vaccination Timing: Evidence from a Discrete Choice Experiment in Wuhan” — ProQuest- Committee: Dohyeong Kim (Chair), Richard Scotch, Soyoung Kwon, Khai Chiong.- Distinction: Dean of Graduate Education Dissertation Research Award (2025). | |
| University of Texas at Dallas | May 2026 |
| M.S. in Social Data Analytics and Research | |
| University of Queensland | 2019 |
| Master of Development Economics | |
| University of California, Riverside | 2017 |
| B.A. in Economics | |

Research Agenda

My research develops behavioral models and AI-driven simulation frameworks to understand why individuals make suboptimal health decisions, and how policy design can be improved through causally-valid computational inference. I integrate Discrete Choice Experiments with structural econometrics and large language models to build Behavioral Digital Twins—empirically-grounded synthetic populations for counterfactual policy simulation. My work spans health economics, behavioral economics, and AI governance.

Working Papers

- “Waiting Time as a Behavioral Barrier to Vaccination Uptake: Nonlinear Heterogeneity by Institutional Trust”**
Under Review at Economic Analysis and Policy — [Link to Full Draft](#)
 - Developed a structural Discrete Choice Experiment (DCE) framework to analyze the interaction between temporal barriers and institutional trust in vaccination behavior.
 - Identified nonlinear heterogeneity in waiting time sensitivity, providing a behavioral explanation for vaccine hesitancy.
- “Empirical-Frontier Regularization for LLM Synthetic Agents: A Preference-Anchored Framework”**
Working Paper (Target: J. Biomedical Informatics) — [SSRN Link](#). Introduces a DCE-grounded empirical-frontier regularization framework for LLM synthetic agents. Using a convex anchoring rule ($\pi = \lambda\pi_{\text{LLM}} + (1 - \lambda)P_{\text{static}}$), the Static-BDT Anchor reduces waiting-time monotonicity violations

from 62.5% to 4.2% and improves Spearman rank correlation from -0.024 to 0.952 . Presented at DAISY 2026 (ACM CHIL 2026) and AIXPH 2026.

3. **“Value Contamination in Policy Simulation: Measuring How LLM Policy-Value Orientations Shape Predicted Administrative Burden Effects”**

Target: Journal of Computational Social Science — [SSRN Link](#). Develops an audit framework for measuring value sensitivity in LLM-generated policy simulations. Using administrative burden in vaccination access as a testbed, evaluates nine LLMs across six model families through policy-value orientation batteries, narrative frame-decomposition, and within-model frame-manipulation experiments.

4. **“CEA-Intismeran: Cost-Effectiveness of mRNA-4157 (Intismeran) + Pembrolizumab vs Pembrolizumab Alone for Adjuvant Melanoma”**

Manuscript under Review at Pharmacoeconomics — [GitHub Repository](#). A cost-effectiveness analysis using partitioned survival modeling (PSM) with parametric hazard extrapolation and Gaussian Process (GP) constraints. Estimates ICER and probability of cost-effectiveness across willingness-to-pay thresholds.

Research Experience & Affiliations

Doctoral Researcher, Spatial Health AI Research Partnership (SHARP), UT Dallas

- Leading interdisciplinary research on the intersection of artificial intelligence, geospatial analytics, and health policy.
- Managing large-scale datasets and developing computational simulations (R, Python) to evaluate pandemic preparedness.
- Leading development of the Behavioral Digital Twin (BDT) framework, integrating DCE data with causally-constrained LLM agents for synthetic policy simulation.

Graduate Researcher, Behavioral Health Economics Laboratory, UT Dallas

- Conducted high-salience behavioral experiments (N=1,027) analyzing risk-weighted health decisions and institutional trust.

Teaching Experience

Teaching Assistant, School of Economic, Political and Policy Sciences, UT Dallas

- **Quantitative Methods for Policy Analysis (Graduate)**: Led Econometrics labs; specialized in R programming, regression analysis, and causal inference.
- **Health Economics & Public Policy (Undergraduate)**: Developed case studies on health market failures and administrative challenges.
- **Public Policy Analysis (Undergraduate)**: Mentored students in drafting evidence-based policy memos.

Conference Presentations

- **AIXPH 2026 (1st Annual Conference on AI and Public Health, Baltimore, MD)**: Empirical-Frontier Regularization for LLM Synthetic Agents — Poster Presentation.
- **APPAM Fall Research Conference 2025** (Seattle, WA): Estimating Time Discount Rate for Vaccination.
- **DAISY 2026 Workshop at ACM Conference on Health, Inference, and Learning (CHIL) 2026**: Behavioral Digital Twins — A Causally-Faithful Generative Framework.

- **UT Dallas Graduate Research Symposium 2025:** Modeling Vaccination Decisions with Hyperbolic Discounting.

Honors & Awards

- **2025** Dean of Graduate Education Dissertation Research Award, UT Dallas
- **2025** Betty & Gifford Johnson Graduate Travel Award, UT Dallas
- **2016** Omicron Delta Epsilon, International Honor Society for Economics

Technical Skills & Certifications

Econometrics: Mixed Logit (MXL), Latent Class Models (LCM), SEIR Modeling, Causal Inference.

AI & ML: PyTorch, Transformers (HuggingFace), LLM APIs (OpenAI, Anthropic), LangChain, Causal Machine Learning.

Software: Stata (Advanced), R, Python, MATLAB, LaTeX, SQL, GIS.

Certifications: Stanford Artificial Intelligence Graduate Certificate; CITI Human Subjects Protection.

REFERENCES

Dohyeong Kim (Chair)

University of Texas at Dallas

Senior Associate Dean of Graduate Education

Robert E. Holmes Jr. Professor of Public Policy,
GIS & Social Data Analytics

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